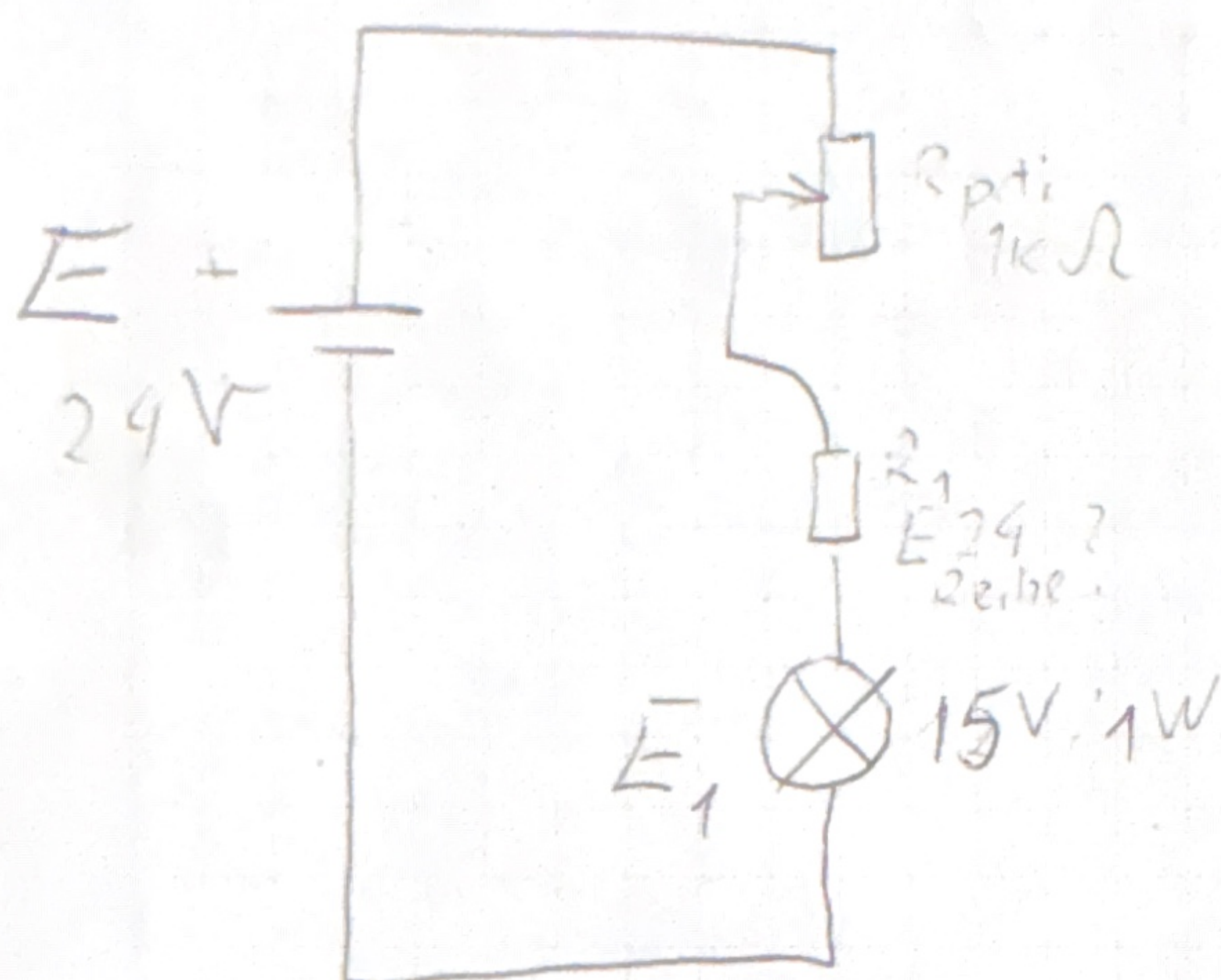




gesucht:

1) R_1

2) P_{R1}

3) P_{pti}

4) Einstellung von P_{pti} (Ein- und Ausschalten) (Lampe leuchtet nicht)

1) $E = I \cdot R_{pti} + I \cdot R_1 + E_1$

$24V = I \cdot 1k\Omega + I \cdot R_1 + 15V$

2) $R_{pti} = 0\Omega$: ~~$24V - 15V = 66,7mA \cdot 1$~~
 $I = \frac{P}{U} = \frac{1W}{15V} = 66,7mA$

$24V = 0 + I \cdot R_1 + 15V$

$24V - 15V = I \cdot R_1$

$9V = 66,7mA \cdot R_1$

$R_1 = \frac{9V}{66,7mA} = 135\Omega$

R_1
 $E_{24} \pm 5\%$

$P_{R1} = I^2 \cdot R_1 = (66,7mA)^2 \cdot 135\Omega$

$P_{R1} = 0,67W$

$\rightarrow R_1 = 150\Omega$

3) $R_{pti} = 1k\Omega$
 $R_{ges} = R_{pti} + R_1 + R_{E1}$

~~$R_{E1} = \frac{1W}{I^2}$~~
 ~~$R_{E1} = \frac{1W}{(66,7mA)^2} = 225\Omega$~~

$R_{ges} = 1k\Omega + 150\Omega + 225\Omega = 1375\Omega$

$P_{E1} = \frac{U^2}{R_{E1}} \rightarrow R_{E1} = \frac{U^2}{P_{E1}} \neq R_{E1} = \frac{225V}{1W} = 225\Omega$

4) $I_{ges} = \frac{E}{R_{ges}} = \frac{24V}{1375\Omega} = 17,45mA$ $P_{R_{pti}} = I_{ges}^2 \cdot R_{pti} = 0,3W$

Ein- und Ausschaltung
5) $E_1 = 0,5W$

$$R_{E1} = 225 \Omega \text{ (Es bleibt so)}$$

$$P = \frac{U_{E1}^2}{R_{E1}}$$

$$U_{E1}^2 = P_{E1} \cdot R_{E1}$$

$$U_{E1} = \sqrt{0,5W \cdot 225 \Omega} = 10,6V$$

$$I_{E1} = \frac{U_{E1}}{R_{E1}} = \frac{10,6V}{225 \Omega} = 47,1mA$$

$$E = I \cdot R_{poti} + I_{E1} \cdot R_1 + E_1$$

$$29V = 47,1mA \cdot R_{poti} + 47,1mA \cdot 130 \Omega + 10,6V$$

$$13,4V = 47,1mA \cdot R_{poti} + 7,065V$$

$$6,335V = 47,1mA \cdot R_{poti}$$

$$R_{poti} = \frac{6,335V}{47,1mA}$$

$$R_{poti} = 134,5 \Omega$$

